

SymPy Bug Card

Bug 19: recurrence solver violates initial conditions

Task. Solve the recurrence with the supplied initial conditions.

$$y(n+2) - (n+2)y(n+1) + y(n) = 0, \quad y(0) = y(1) = 1.$$

SymPy's answer. SymPy's `rsolve` returns the zero sequence as the solution.

Correct answer. A valid answer must satisfy $y(0) = 1$ and $y(1) = 1$, or the solver should report that it cannot produce such a solution; the zero sequence is not a correct answer.

Explanation. Substituting into SymPy's returned expression gives $0|_{n=0} = 0$ and $0|_{n=1} = 0$, contradicting both supplied initial conditions. Forward recurrence from the requested values starts $1, 1, 1, 2, 7, 33, \dots$, so this is not an equivalent representation of the requested initial-value solution.

Likely root cause. The diagnosis locates the issue in `sympy/solvers/recr.py:rsolve:` initial conditions are applied only when the internal solution contains arbitrary constants. Here `rsolve_hyper` returns the constant-free candidate 0 with no symbols, so validation is skipped. This is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-019-recurrence-solver-violates-initial-conditions/](#)